

Challenge (Habanero)

- Q1.** A hiker travels 24.5 miles in a day. What is the remainder when this distance is divided by 9?
- (A) 0
(B) 3
(C) 6
(D) 9
(E) 12
- Q2.** Which number is divisible by 2 and 5 but not by 9?
- (A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q3.** A park ranger tracks the population of deer. If the population increases by 150 deer per year, what is the remainder when the total population after 7 years is divided by 10?
- (A) 0
(B) 2
(C) 5
(D) 7
(E) 9
- Q4.** What is the result of $48 \div 6$ with remainder?
- (A) 8R0
(A) 7R4
(A) 6R6
(A) 8R2
(A) 6R0
- Q5.** A fuel tank can hold 240 gallons. If 120 gallons are already filled, what fraction of the tank is filled?
- (A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{2}{3}$
(D) $\frac{3}{4}$
(E) $\frac{1}{1}$
- Q6.** A mountain trail is 54 km long. What is the remainder when the length of the trail is divided by 9?
- (A) 0
(B) 3
(C) 6
(D) 9
(E) 12
- Q7.** If a bird sanctuary has 135 birds and 15 more birds arrive, what is the remainder when the new total is divided by 10?
- (A) 0
(B) 2
(C) 5
(D) 7
(E) 9
- Q8.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, what is the remainder when the candy is divided?
- (A) 0
(B) 2
(C) 4
(D) 6
(E) 8
- Q9.** A car travels 225 miles on a full tank of gas. If the gas tank holds 15 gallons, how many gallons of gas are used per mile? Show your work and express your answer as a fraction or decimal.
- Q10.** A conservationist is tracking the growth of a wolf population. If the population doubles every 5 years and there are currently 18 wolves, how many wolves will there be in 15 years? Show your work and express your answer as an integer.

Open Ended Questions (FRQ)

- Q9.** A car travels 225 miles on a full tank of gas. If the gas tank holds 15 gallons, how many gallons of gas are used per mile? Show your work and express your answer as a fraction or decimal.
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Chef's Tips

- Read each question carefully
- Use the back of the page for calculations if needed
- Show all work for free response questions